Local logistic regression for dimension reduction in classification

Touqeer Ahmad[1,*], François Portier[1], and Gilles Stupfler[2]

[1] CREST, Ensai, France. touqeer.ahmad@ensai.fr, *(Presenting author)
[2] Université d'Angers, France.

Outline

This work explores binary classification challenges in high-dimensional data. It introduces a penalized local logistic regression for dimension reduction as a crucial solution to mitigate the adverse effects of high-dimensional feature space.

Introduction

In standard classification problems, when dealing with high-dimensional data, it is essential to reduce the dimensionality of the original predictor space before conducting statistical analysis. A large number of features can significantly affect statistical analysis tools, leading to challenges in model interpretability.

Background

Let $Y \in \{0,1\}$ and $X \in \mathbb{R}^p$ and let P denote the probability measure of Y, X. We want to reduce the dimension of X while still keeping the same information to predict Y. According to (Cook, 1996), the sufficient dimension reduction (SDR) follows the condition that

$$Y \perp X | P_\beta X,$$

where $P_\beta = \beta\beta^T$ is a projection on $\operatorname{span}(\beta)$, a (possibly) small dimension subspace of $\mathbb{R}^p$. Equivalently, it means that

$$\mathbb{P}(Y=1|X) = \mathbb{P}(Y=1|\beta^T X)$$

Set $d = \dim(P_\beta)$. When $d = p$, then no dimension reduction is expected and the previous condition does not imply.

How model works

Let $Y \in \{0,1\}$ be a binary response variable, with random covariate $X \in \mathbb{R}^p$. Suppose that for an x in the support S_X of X (assumed to have nonempty interior),

$$\pi(x) = \mathbb{P}(Y=1|X=x) = g(\beta^T X),$$

for some unknown $\beta \in \mathbb{R}^{p\times d}$, $d \le p$ and an unknown measurable function $g : \mathbb{R}^d \to [0,1]$. When $g = g_0$ is known, the integrated cross entropy-based loss function

$$-\mathbb{E}[Y \log(g_0(b^T X)) + (1-Y)\log(1-g_0(b^T X))]$$

viewed as a function of b. This naturally suggests the estimator

$$\underset{b\in\mathbb{R}^{p\times d}}{\arg\min} -\frac{1}{n}\sum_{i=1}^{n} Y_i \log(g_0(b^T X_i)) + (1-Y_i)\log(1-g_0(b^T X_i)).$$

For $d = 1$ and $g_0 = \operatorname{expit}$, this is just the standard logistic regression estimator.

However, if π is differentiable at x, then the gradient $\nabla\pi(x)$ of π at x satisfies

$$\nabla\pi(x) = \beta\nabla g(\beta^T x).$$

As such, $\nabla\pi(x) \in \operatorname{span}(\beta)$. If moreover $0 < \pi(x) < 1$, then likewise

$$\nabla\ell(x) = -\frac{1}{\pi(x)(1-\pi(x))}\nabla\pi(x) \in \operatorname{span}(\beta) \text{ where } \ell = \operatorname{logit}\pi.$$

To recover $\operatorname{span}(\beta)$, it then suffices to estimate enough gradients of the form $\nabla\ell(x_j)$. Now a Taylor expansion of p around x yields

$$\pi(y) \approx \operatorname{expit}(a + b^T(y-x)),$$

with $a = \operatorname{logit}(\pi(x))$ and $b = \nabla \operatorname{logit}(\pi(x)) = \nabla\ell(x)$.

This suggests that the logistic can be used to produce an estimator of $\nabla\ell(x)$ if it is restricted to data points close enough to x; to put it differently, one should expect the local integrated cross-entropy

$$\begin{aligned} &-\mathbb{E}[Y\log(\operatorname{expit}(a + b^T(X-x))) \\ &+ (1-Y)\log(1-\operatorname{expit}(a+b^T(X-x))) \,|\, X \in B(x,\varepsilon)] \end{aligned} \tag{1}$$

to be minimized at a pair $(a_\varepsilon(x), b_\varepsilon(x)) \approx (\ell(x), \nabla\ell(x))$ as $\varepsilon \downarrow 0$, where $B(x,\varepsilon)$ is the closed Euclidean ball with radius ε in $\mathbb{R}^p$.

This is the cornerstone of the methodology we developed.

Nearest-neighbor penalized local logistic estimator

- The localization induced upon covariate empirically using a nearest-neighbor procedure.
- Fix $x \in \mathbb{R}^p$ and let $N_k(x) \subset \{1,\dots,n\}$ be the set gathering the indices of the k-nearest neighbors X_i of the point x.

The empirical counterpart of the local integrated cross-entropy is then

$$\begin{aligned} &-\frac{1}{k}\sum_{i\in N_k(x)} Y_i \log(\operatorname{expit}(a + b^T(X_i - x))) \\ &+(1-Y_i)\log(1-\operatorname{expit}(a+b^T(X_i-x))). \end{aligned} \tag{2}$$

Introducing the LASSO penalty $\lambda\|b\|_1$, where $\|\cdot\|_1$ is the ℓ_1−norm on $\mathbb{R}^p$, and rescaling, we naturally obtain a nearest-neighbor, penalized, local logistic estimator as

$$(\hat\alpha_n(x), \hat\beta_n(x)) = \underset{(a,b)\in\mathbb{R}\times\mathbb{R}^p}{\arg\max}\{L_n(a,b) - \lambda\|b\|_1\} \tag{3}$$

with

$$\begin{aligned} L_n(a,b) = &\sum_{i\in N_k(x)} Y_i\log(\operatorname{expit}(a+b^T(X_i-x))) \\ &+(1-Y_i)\log(1-\operatorname{expit}(a+b^T(X_i-x))). \end{aligned}$$

Aggregating the directions

The optimization procedure (3) allows to estimate $\nabla_x\ell(x)$ which belongs to the central subspace for each $x \in S_X$.

$$M = \int_{\mathbb{R}^p}\nabla_x\ell(x)\nabla_x\ell(x)^T\hat\mu(dx) = \mathbb{E}_{X^*\sim\hat\mu}[\nabla_x\ell(X^*)\nabla_x\ell(X^*)^T].$$

To estimate the matrix M, we first generate $X_i^* \sim \hat\mu$, $i = 1,\dots,m$ independently. Second, we compute the empirical average

$$\widehat{M} = \frac{1}{m}\sum_{i=1}^{m}\hat\beta(X_i^*)\hat\beta(X_i^*)^T. \tag{4}$$

Once this estimator $\widehat{M}$ of M has been calculated, one can define $(\hat\beta_1,\dots,\hat\beta_p)$ as the set of eigenvectors of $\widehat{M}$, ordered according to their eigenvalues (in decreasing order).

Algorithm

In order to estimate M, we use algorithm 1.

Algorithm 1 Algorithm developed for estimation of M.

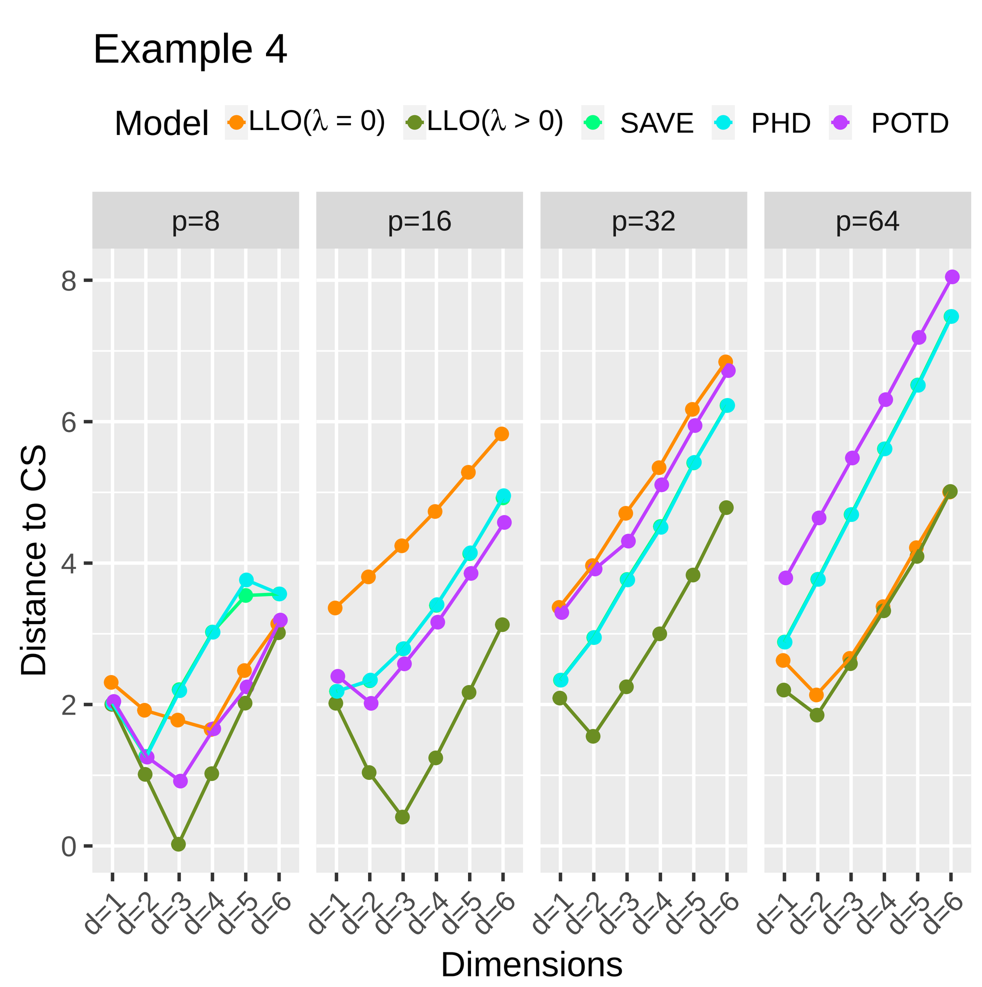
Input: $(X_1,Y_1),\dots,(X_n,Y_n) \subset \mathbb{R}^p\times\{0,1\}$, $\lambda > 0$, $k \in \{1,\dots,n\}$ and $m \in \{1,\dots,n\}$.
Output: Dimension reduction matrix $\widehat{M}$
Draw uniformly a list $\mathcal{X}_m$ of m observations among $\mathcal{X} = (X_1,\dots,X_n)$ without replacement
for each $x \in \mathcal{X}_m$ **do**
  Compute $N_k(x)$, the index of the k nearest neighbors to x among $\mathcal{X}$.
  Compute $\hat\alpha_n(x)$ and $\hat\beta_n(x)$ according to (3)
end for.
Return $\widehat{M} = \frac{1}{m}\sum_{x\in\mathcal{X}_m}\hat\beta_n(x)\hat\beta_n(x)^T$.

Synthetic data example

Example 4. We generated a binary response variable as

$$Y = \operatorname{sign}\left\{\log(X_1^2)(X_2^2 + X_3) + 0.2\epsilon\right\}.$$

where $X, \epsilon \sim \mathcal{N}(0,1)$. For fixed $n = 1000$



Real data appliaction

The real application considers the Hill-Valley (HV) data set leading to 100 features in total and the binary nature response variable represents vally=0, hill=1. We apply our developed **knn** based cross-validation forward selection procedure.

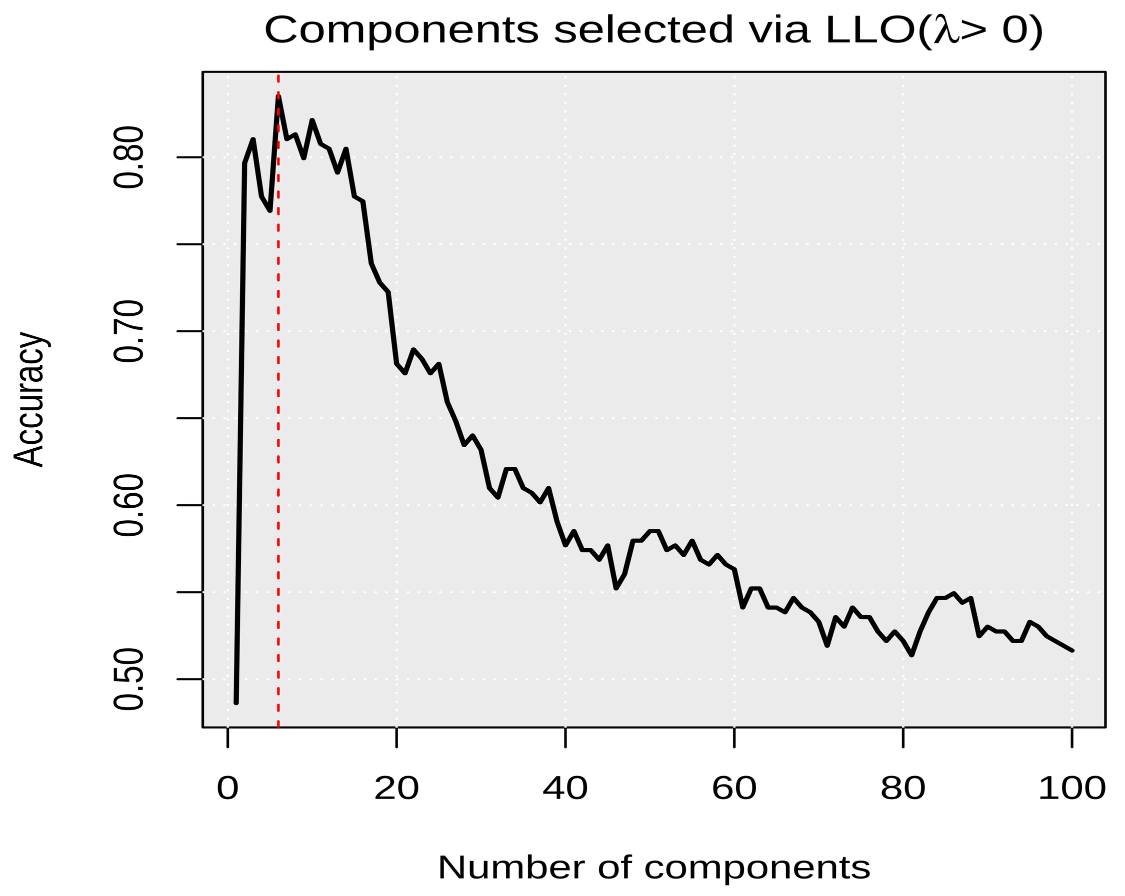


Figure 2. Important dimension selection through cross-validation for LLO($\lambda > 0$)

The classification rule is learned on the training set through **knn** classifier. Table 1 provides the assessment measures results.

Table 1. Misclassification risk, the area under the curve, and computing time of the models when using **knn** classifier.

Model	Miscl risk	AUC	Est time
No DR	0.481	0.516	1.48
LLO ($\lambda = 0$)	0.429	0.597	0.73
LLO ($\lambda > 0$)	0.126	0.953	0.73

Compression with existing methods

The Misclassification risk in Figure 3 reveals that the proposed LLO($\lambda > 0$) outperforms than existing competitors.

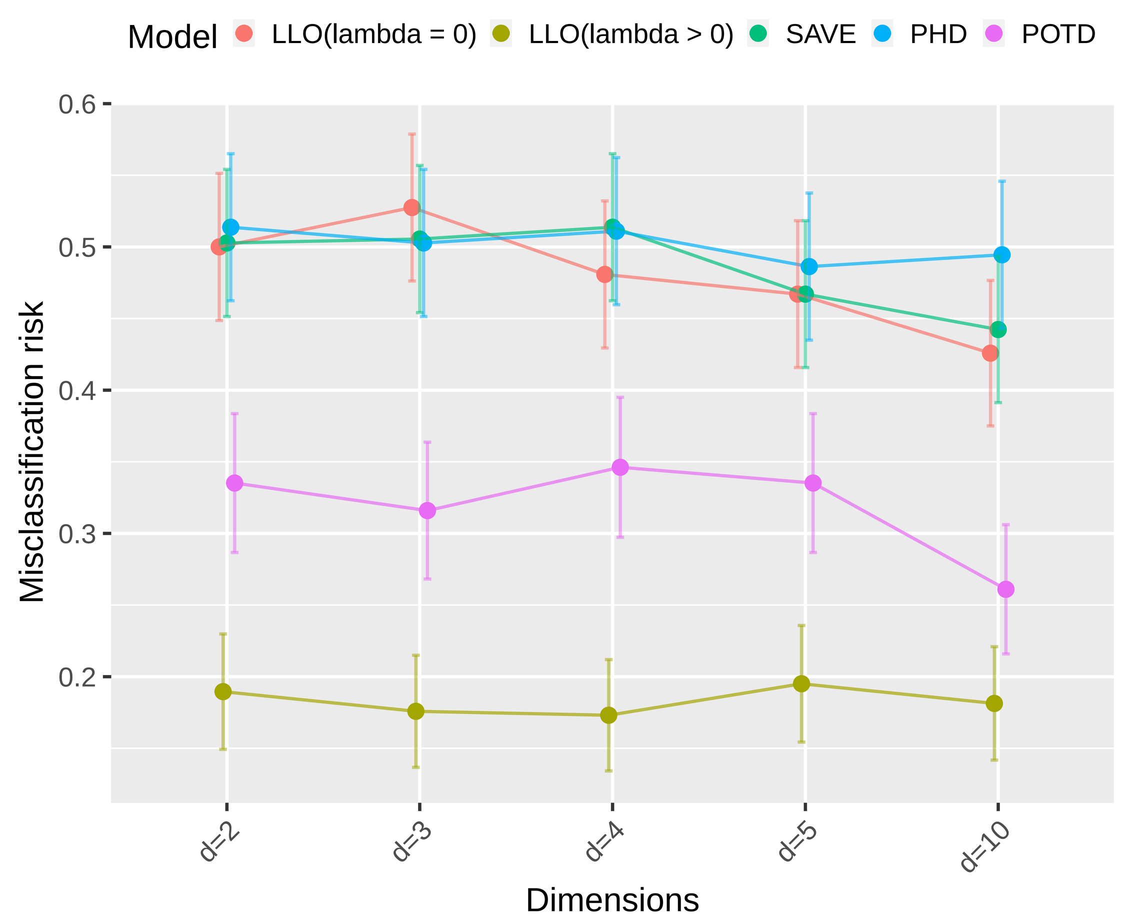


Figure 3. Misclassification risk for proposed and existing models.

Ongoing work

- Dimension reduction for the tail event (Aghbalou et al. 2024).

$$\pi(t|x) = \mathbb{P}(Y > t | X = x)$$

where t is a sufficiently large threshold that defines the tail events

References

Cook, R. D. (1996). Graphics for Regressions With a Binary Response. *Journal of the American Statistical Association, 91(435)*, 983–992. https://doi.org/10.2307/2291717

Aghbalou, A., Portier, F., Sabourin, A., & Zhou, C. (2024). Tail Inverse Regression: dimension reduction for prediction of extremes. *Bernoulli*, 30(1), 503-533.